# Session 04 (AIML) - Assignment Questions

**Q1.** (Lambda Function)
Write a lambda function named square that takes a number x and returns its square.
Call the lambda function with input from the user and print the result.
Also write the same logic using a normal def function and compare both in comments.

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**Q2.** (Lambda with Multiple Arguments)
Create a lambda function add_three(a, b, c) that returns the sum of three numbers.
Take three numbers as input from the user and print their sum using this lambda function.

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**Q3.** (For Loop Revision)
Write a program using a for loop and range() to:
a) Print all numbers from 1 to 25.
b) Print the sum of all numbers from 1 to 20.
c) Print the table of 5 (5 × 1 to 5 × 10).

Add comments explaining when to use for loop.

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**Q4.** (While Loop Revision)
Write a program using a while loop that:

- Repeatedly asks the user to enter a positive number.
- Keeps adding the numbers to a running total.
- Stops when the user enters 0 or a negative number.
- Finally prints the total sum.

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**Q5.** (Math Module)
Write a program that imports the math module and does the following:

- Takes a number as input.

- Prints its square root using math.sqrt().
- Prints its factorial using math.factorial() (for positive integers only).
- Prints the ceiling and floor value using math.ceil() and math.floor().

Add comments about the usefulness of the math module.

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**Q6.** (Random Module)
Write a program using the random module that:

- Generates and prints 5 random integers between 1 and 100.
- Generates and prints one random number between 50 and 150.
- Prints a random floating point number between 0 and 1.

Use different functions from the random module.

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**Q7.** (Import Methods)
Demonstrate different ways of importing modules:

1. Import the entire math module using import math and use math.pow(2, 4).
2. Import only sqrt from math using from math import sqrt.
3. Import math with an alias as m and use m.factorial(5).

Print the results with clear messages.

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**Q8.** (Custom Module)
Create a custom module file named mymodule.py that contains:

- A function greet(name) that prints "Hello [name], Welcome to Python Class!"
- A function calculate_power(base, exp) that returns base raised to exponent.

Then write a main program that imports this module and uses both functions.

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**Q9.** (**name** == "**main**")
Write a program that:
a) Creates a small function and demonstrates the use of if __name__ ==
"__main__": so that some code runs only when the file is executed directly.

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**Q10.** (Mini Project - Combined Concepts)
Create a **Simple Math Utility Program** using functions, lambda, loops and modules:

Write the following:

1. A lambda function to calculate square of a number.
2. A normal function using math module to calculate power.
3. Use a while loop to repeatedly ask the user for a number and choice (Square / Power / Random Number).
4. Use random module to generate a random number when chosen.

Continue until the user chooses to exit.
Use proper functions and comments.